

SOUBHIK ROY

Dhanbad, Jharkhand

☎ +91-9835518101 ✉ soubhik0727@gmail.com 🌐 Portfolio 🔗 LinkedIn 🐙 GitHub 📄 LeetCode

EDUCATION

Vellore Institute of Technology, Bhopal
B.Tech - Computer Science and Engineering - CGPA - 8.2

Sep 2023 – May 2027
Bhopal, Madhya Pradesh

EXPERIENCE

The 24 Fitness Gym **Oct 2025 – Jan 2026**

Full-Stack Developer *Remote*

- Built a production-ready gym management platform using Next.js, TypeScript, and Supabase, supporting **3 roles** (User, Trainer, Admin) and **15+** core workflows with RBAC and full lifecycle control.
- Implemented end-to-end membership and trainer lifecycle (purchase, renewal, **15-day GP**, auto-expiry), reducing manual admin operations by **65%** and preventing data inconsistency.
- Developed secure payment verification, PDF invoice generation, real-time messaging, and transactional emails (Resend), handling **100+** transactions/month with **99%** delivery reliability and **75%** higher user engagement.

IIT (ISM) Dhanbad **May 2025 – June 2025**

Summer Research Intern *Dhanbad, IN*

- Engineered EV-routing algos, cutting TAT by **20%** and increasing vehicle load efficiency by **30%** across 500+ km routes for a 20-truck fleet.
- Designed and executed **50+** automated test cases, improving test coverage to **95%** and reducing regression testing time by **40%**.
- Collaborated with research mentors to analyze real-world logistics data and validate algorithm performance, improving route feasibility and deployment readiness by **35%**.

TECHNICAL PROJECTS

EV Routing & Charging Optimization — Python, Flask, Genetic Algorithms **May 2025 – July 2025**

- Developed an EV routing algorithm optimizing delivery routes, charging schedules, and cost across 15+ customer nodes.
- Implemented multiple delivery modes (**fastest**, **min_cost**, **balanced**) using graph-based heuristics.
- Built a **Flask REST API** with real-time route visualization and benchmarking dashboards.
- Achieved **23% route efficiency improvement** through partial charging and multi-EV dispatching strategies.

AI Model Benchmarking Tool — Next.js, PostgreSQL **Feb 2025 – Apr 2025**

- Evaluated over **25 ML models**, focusing on **F1-score optimization (72%)**, achieving **latency <50ms**, **memory usage <100MB**, and **throughput >1,000 req/s**.
- Automated PDF-based eval reports, reducing manual analysis time by **75%** for **200+ users**.
- Deployed with **Supabase**, ensuring **99%+ uptime** under **150+ concurrent user** load.
- Accelerated batch data ops by **65%** using Pandas vectorization (15min → 5.2min).

PUBLICATIONS

AI Meets Load Balancing — Research Lead **June 2024 – Oct 2024**

- Applied **Reinforcement Learning** in **K8s/Envoy** stack, achieving **99.9% system uptime** and reducing manual ops by **80%**, saving **200+ dev hours/month**.
- Engineered traffic-aware model selector reducing latency by **30%** (210ms → 147ms) across **50+ microservices**, scaling to **5K+ RPS**.
- Formulated dynamic auto-scale policies, improving infra utilization by **40%** and maintaining **<1% request failure rate**.

ACHIEVEMENTS & CERTIFICATIONS

Hackathon: 3× Runner-up
Problem Solving: GeeksforGeeks – 160 Days of DSA (04/2025)
Frontend Dev: Infosys Springboard (9/2024)
Java Certified – Udemy (6/2024)

TECHNICAL SKILLS

Languages: Java (Spring Boot), Python (Pandas, Flask), TypeScript
Frameworks: Next.js, React.js, Node.js, Express.js
ML/AI: TensorFlow, PyTorch, Scikit-learn (Model Optimization, F1-score 72%)
Databases: PostgreSQL, MongoDB (50+ transactions/day)
DevOps/Cloud: AWS (EC2, S3), Docker, CI/CD Pipelines (Jenkins, GitHub Actions)